QUESTION: HOW CLOSE CAN ONE POLICY DESIGN GET TO PERFECT RELIABILITY ACROSS STATES AND INDEPENDENTLY TRAINED ACTORS?

Supervision nearly closes the observed reliability tail. Pure RL still finds better trajectories.

Across 12,505 seed-state trials, the selected mixed deployment has nine near-reference failures, compared with 673 for selected pure RL. Pure RL wins strictly against the stored reference more often.

99.928%
RL + supervised near-reference success
12,496 / 12,505 trials
Task success 93.858% · strict wins 12.555%

94.618%
pure-RL near-reference success
11,832 / 12,505 trials
Task success 92.499% · strict wins 18.417%

Task and return
Pendulum, 200 deterministic steps. $r = -(\theta^2 + 0.1\theta^2 + 0.001a^2)$. Torque a is in [−2, 2].

Near reference
$R_\pi \geq R^* - 5$
R^* is the better stored dynamic-programming or controller return.

Task success
$\cos\theta \geq 0.95$ and $|\theta| \leq 1$ for at least 80% of steps; no failure streak longer than 50 steps.

Strict win and grid
$R_\pi > R^*$. 61 angles × 41 velocities × actor seeds 0–4. Grid trials are correlated.

1 · Method

Two routes, no reference at deployment

Mixed, RL + supervised

1. TRAIN A SimbaV2 actor, with normalized residual network blocks, learns reference actions. Three DAgger rounds then request reference labels on 30,000 states visited by the learner.
2. TRAIN A 20,000-state follow-up automatically prioritizes starts where the frozen actor performs worst.
3. DEPLOY A reward-only FastSACN8 critic, trained with one-step and eight-step return targets, checks the actor action and four nearby torques. Both critics must approve a change.

Pure RL

1. TRAIN SimbaV2 SAC, with normalized residual network blocks, learns from reward for 100,000 transitions.
2. DEPLOY Reflection averages the actor with its sign-flipped state and action counterpart: $a_{sym}(s) = [\pi(s) - \pi(Ms)] / 2$.
3. DEPLOY Search 41 torques in [−2, 2]. Replace a_{sym} only when both critics predict an improvement above 0.005.

Selected-route ledger

Method	Learn	Discover	Labels
Mixed	100k	800k	690k
Pure RL	100k	0	0

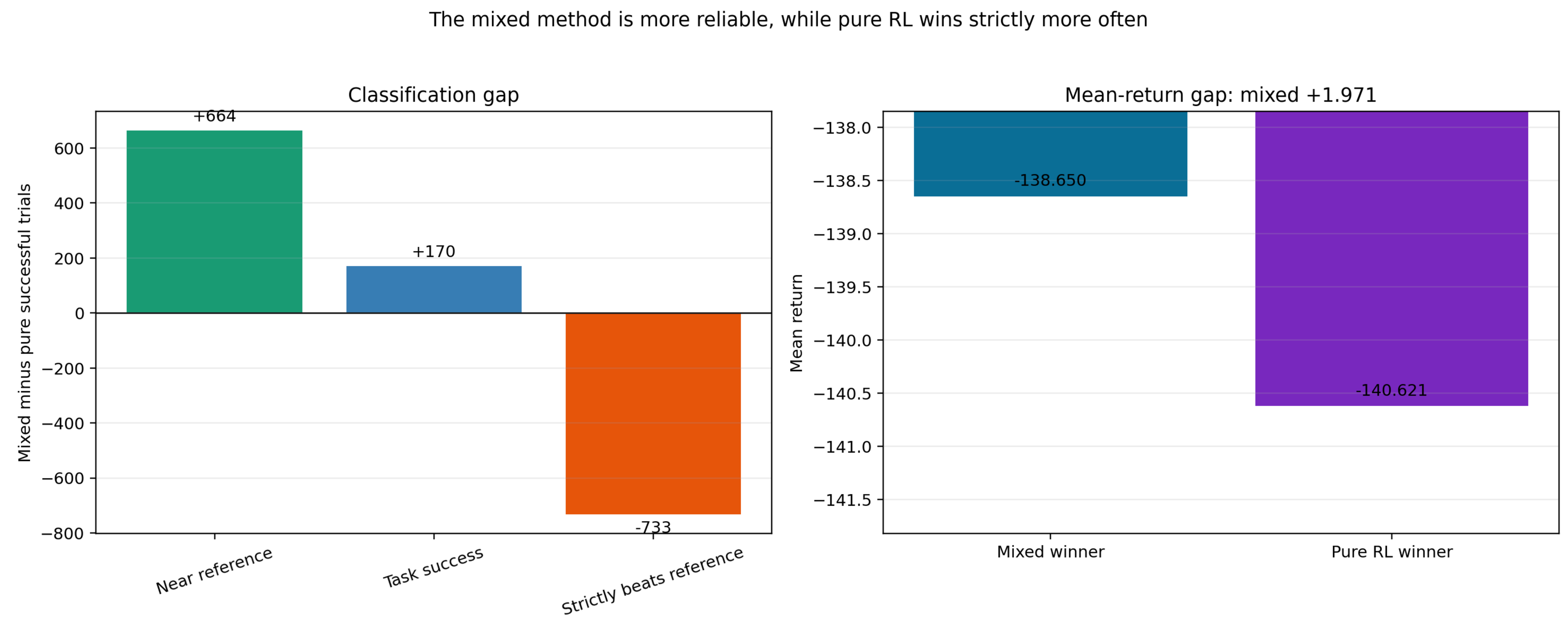
Units: Mixed Learn = 30k + 20k actor-state transitions and one shared 50k critic; pure Learn = 100k reward transitions. These learning events are not compute-equivalent. Discover = 800k candidate rollout interactions used only to rank starts. Labels = 400k initializer + 30k learner states + 240k labels from uniformly sampled reset states + 20k priority states.
Sampler footnote: the mixed initializer oversampled near-upright states within 35°. Its deployed actor and Q-search use no angle rule or reference query.

2 · Primary evidence

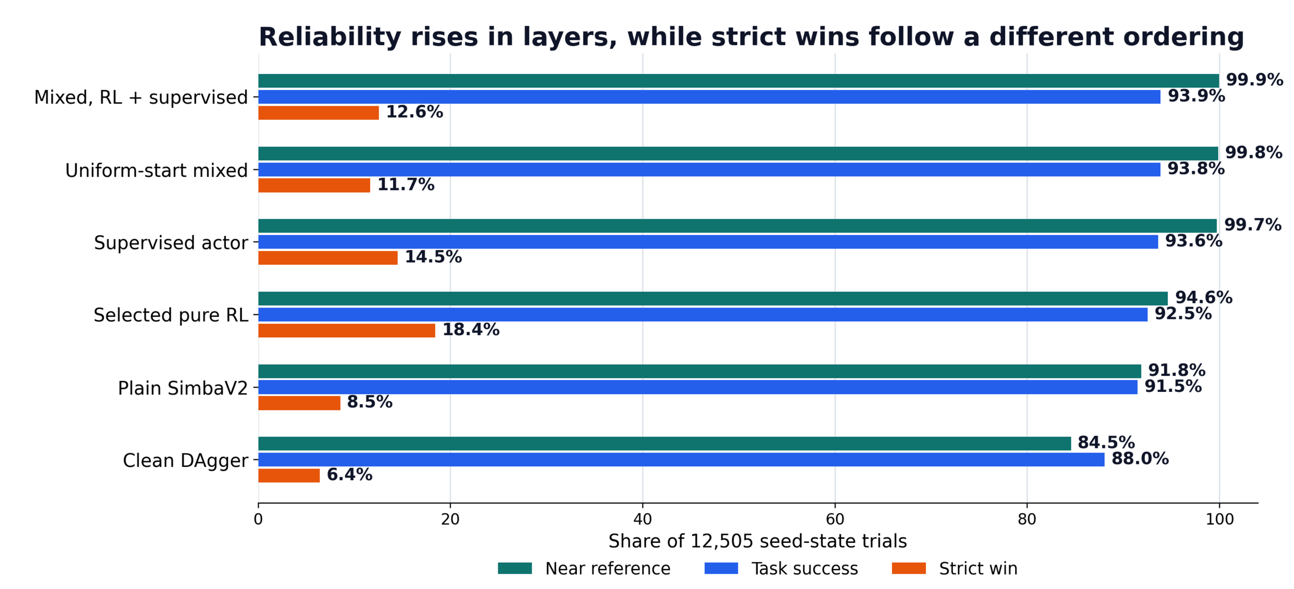
Nine deployed failures versus 673

The nine mixed failures occur across eight of 2,501 grid cells. These are retrospectively selected five-seed grid results from 404 standardized evaluations, not a held-out confirmation set. Pure RL discovers more trajectories that exceed the stored comparator.

How large is the category gap?



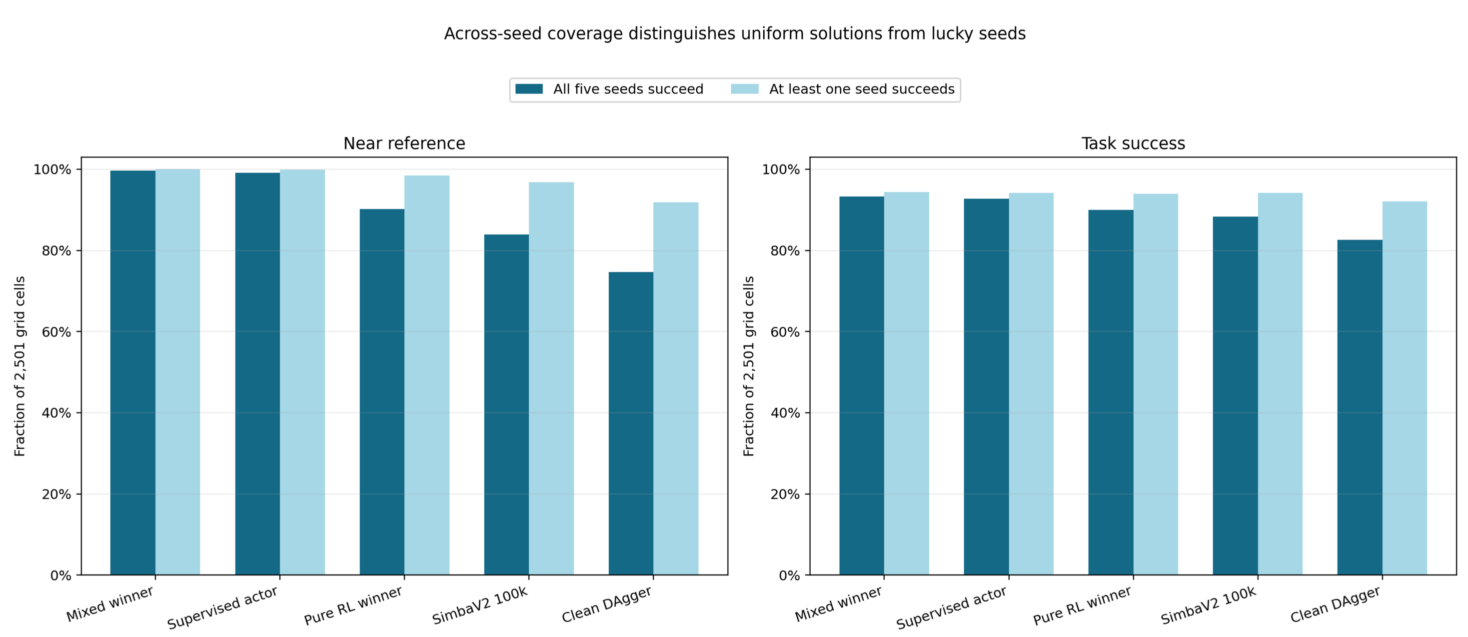
How do the main baselines compare?



3 · Seed diagnosis

The gap is a consistency gap across seeds

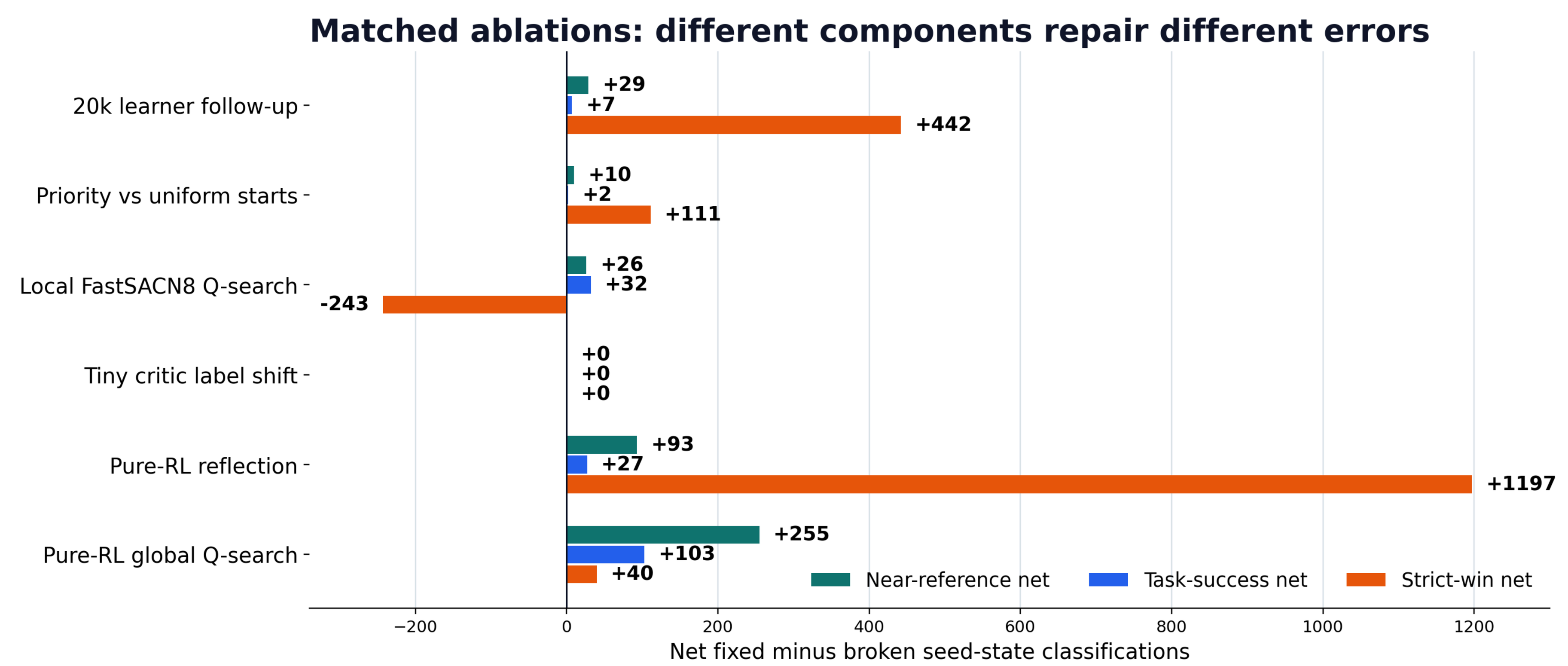
All five mixed actors succeed on 2,493 of 2,501 cells; per-seed rates span 99.72% to 100%. Pure RL reaches 2,255 all-seed cells, spans 93.32% to 96.24%, and has 207 seed-variable cells.



4 · Matched ablations

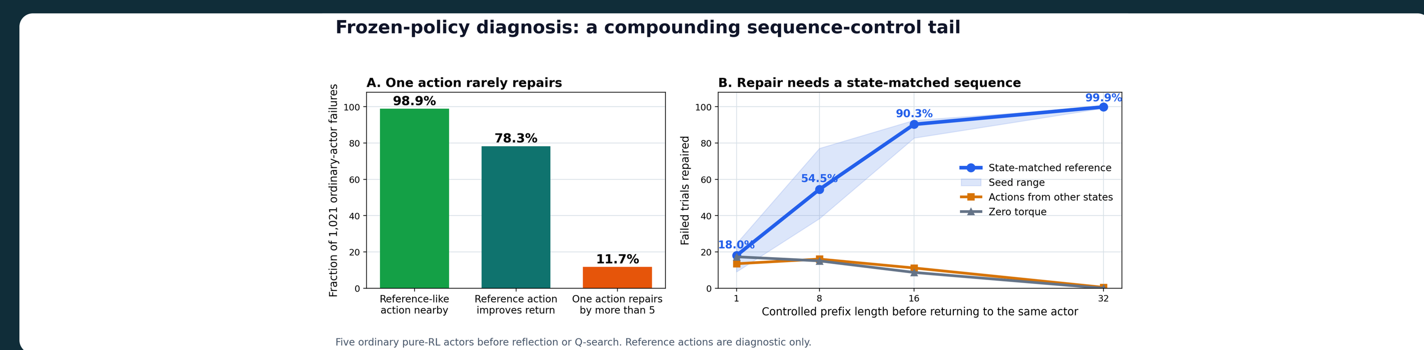
Different components repair different errors

Bars show after minus before on the same 12,505 trials. In a separate locked off-grid combination test, FastSACN8 plus critic UTD2 lowers the raw actor from 91.86% to 89.94%, but the same reflection and Q-search raise 94.49% to 95.91%. It loses 25 task-success trials and misses the promotion gate.



5 · Mechanism diagnostic

The pure tail is not a simple coverage hole



98.9% nearby
A reference-like action already occurs among the 256 nearest replay actions for 1,010 of 1,021 failed pre-intervention actor trials.

11.7% repaired
One reference action produces a return gain above five in only 119 failures.

99.9% at 32 steps
A state-matched reference prefix repairs almost every failed trial before control returns to the same frozen actor.

0.5% shuffled
Reference actions taken from other states do not reproduce the repair. Zero torque repairs none at 32 steps.

Scope: these 1,021 failures come from five frozen pure-RL actors before reflection or Q-search. The 673 headline failures come after both deployment corrections. Reference actions are diagnostic only.

Failed seeds do not have farther nearest training resets more often than chance, and useful nearby replay actions are usually present. One action rarely repairs the return, while a 32-step state-matched sequence repairs 99.9%. The evidence points to compounding sequence error and imperfect actor-critic transfer without identifying the training cause.

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SAC: Haarnoja et al. 2018 · DAgger: Ross et al. 2011 · SimbaV2: Lee et al. 2025 · SACn: Łyskawa et al. 2025

61 angles × 41 velocities × 5 actor seeds